Dissertation and torque-paper audit

30 October 2026. Findings only. Three subagents audited torque/paper/supplement, SADb/Appendix E, and SNS/Discussion/Appendix D. The lead reviewed source authority, prior comments/reports, Appendices A-C, build results, and the combined findings. No manuscript, figure, bibliography, data, model, or Overleaf review state was edited.

Verdict

The dissertation contains substantial, defensible doctoral engineering contributions: experimentally informed actuator and torque models, an application linking physiological evidence, new simulation interfaces, and controlled verification experiments. The report does not grade Ben or infer his ability from his advisor's tone. The remaining risks are specific documentation contradictions, difficult-to-read evidence, and repeated prose. These merit a focused reconciliation and presentation pass. They do not establish a need to repeat the optimizations or demonstrate harness-free walking.

The torque paper is not yet submission-cleared. Its scientific contribution and hardware evidence are present, but its sign/margin definitions, coordinate inventories, bibliographic metadata, and final-size plots need attention before circulation. Some issues are confirmed textual contradictions; others explicitly require checking an existing saved definition before deciding whether any value is wrong.

Authority and verification

Item	Reviewed authority / outcome
Dissertation	`Bolen_Dissertation.zip`, saved 29 Oct23:52, SHA-256 in SOURCE_MANIFEST.json; isolated under Archive/audit_20261030/dissertation.
Torque paper	`Design_Factors_for_Isometric_Robotic_Knee_Torque.zip`, saved 29 Oct23:53; separate isolated paper snapshot.
Fresh dissertation build	306 pages, zero TeX errors or undefined/missing references; one inherited 0.66867pt overfull box.
Fresh paper/supplement builds	33+10 pages, zero TeX errors or undefined references after cross-reference convergence. Frontiers font/box warnings remain. Supplement has no independent bibliography; an unnecessary BibTeX invocation returned1 for absent citation/style commands. That is not a supplementary manuscript failure.
ZIP's uploaded PDFs	Paper ZIP contains stale 38-page main/five-page supplement PDFs. The current source builds 33/10. This does not prove that Overleaf's current compiled preview is stale.
Local correspondence	ZIP Chapter 4/5/7 and Appendices A-E match local sources after newline normalization. Chapter 6 has a same-length local difference; the ZIP is the audit authority. Figures were inspected from the ZIP, so older local asset status is not assumed.

Comments and tracked changes are not contained in a normal Overleaf source ZIP. This audit re-read the saved 42-card Chapter 4/5 ledger, Ben's plain-text Word instructions, the newly captured 29 Oct reply, CHATGPT_REPORT.md's relevant review/progress records, and the consolidation/flow/slimming/upload reports. The overnight ZCode report records tracked-change acceptance and later source uploads; that is a historical report, not an independently verified present count. Live review/history inspection could not be completed because desktop automation repeatedly lost target-window geometry. Consequently this report does not claim a complete new live-comment census or count remaining tracked changes. It does not mark any thread resolved.

The original comments teach concrete rules: result before figure; topic sentences; explicit objects instead of unexplained 'views' or 'following'; preserve deletions; separate evidence from model assumptions; and demonstrate the curated resource with a relevant readable record. Ben's rulings take precedence over inferred requests. In particular, FVU is closed to further revision, and the database classification sweep remains deferred. The AI-tool comparison is

excluded from this audit under Ben's clarification; no documented-method or wording task remains. Earlier reports that called such a method 'requested' were incorrect and are superseded here.

First repair priorities

Each row identifies an evidence-backed issue and its smallest useful response. P1 means a correctness, interpretation, reproducibility, or distribution issue that blocks clearance until reconciled. P2 is substantive reader/presentation repair. P3 is optional polish. These are editorial priorities, not institutional degree criteria.

Priority / detail	Finding	Minimal response
P1 A1	Appendix C says the identified flexor uses two-rotation frames, then calls the same implementation pitch-only; Appendix A and Table C9 repeat the latter.	Align saved-fit versus current-evaluator descriptions using existing provenance. Do not guess a convention or refit.
P1 T01	Cam margins over1.05-times-human and over human do not obey the written relationship if defined on the same domain.	Identify saved definitions/targets/domains, then correct only a verified definition or transcription.
P1 T02	Paper introduces signed torque as rF while defining r and F as nonnegative magnitudes.	Use the magnitude equation, as Chapter 4 already does, or define signed moment arm.
P1 T03	Supplement supplies two coordinate sets for a named biomimetic flexor without a frame/version mapping.	Identify the configuration and transform for each table; make one inventory authoritative.
P1 SNS-S01	The gait-score description lacks a versioned operational definition; current code's offset term/bonus conflicts with the simple 'zero perfect/mean-removed' description.	Map each archived result to its existing scorer; define terms and sentinel. Current-code differences are conditional evidence, not proof all historical scores are wrong.
P1 T04	Uploaded PDFs bundled in the paper ZIP disagree with its source.	Exclude or replace stale generated PDFs in the distributed submission package.
P2 SADb-S1	The advisor's criticized Table screenshot still foregrounds Deep learning/turbulence/archived records.	Use a genuine relevant current filtered capture in the main chapter, preserving archive access.
P2 SADb-S2	Appendix E's38 'without grounding text' counts empty notes, not proven absent source text.	State exactly what the saved field counts; distinguish note coverage, source availability, and archive status. No corpus review is required to correct the label.
P2 SNS-S02/S03	Contact duty is described as cycle-based for a leg with zero detected cycles; an old startup comparison is called settled/matched without identifying its relation to the new transient configuration.	Correct captions from existing diagnostic definitions; label old startup versus new transient protocols.
P2 T05/T11	The paper calls the published force-model paper unpublished; its author list differs from the dissertation's publication note.	Correct published metadata; have the human authors settle the author list/title.
P2 T06/T07/A2	Paper retains the removed hybrid series; several legends are 5-6pt at final size; Figure B1 caption uses six letters while its rows use A/B/C.	Reconcile paper carryover deliberately; enlarge useful labels and fix caption letters. Preserve all actual data.
P2 SNS-S04	Figure 7.1 sends closed-loop verification straight to hardware, while prose requires a second acceptance check.	Add the missing decision/return route or narrow the prose.

The margin, scorer-version, and frame issues must be resolved from existing records before numerical edits. None authorizes new optimizations, calibration, or experiments.

Prose and page design

Scope	Current state	Best targeted improvement
Chapter 4	The new direct-route/cam tables, claim-first paragraphs, 1x2 flexor figure, and Exceeding Human Torque Magnitude section improve the story.	Compress equivalent compliance substitutions while preserving the advisor-requested mechanics; restore axis units/frame on Figure 4.12.
Chapter 5	Four tabs, embedded/offline distinction, citation directions, active curation, and callout terminology are now clear.	Trim repeated corpus/field counts; move the useful app demonstration closer to first discussion. Printed119-123 are text before the map on124. Crop/recompose the Pivot screenshot rather than adding another figure.
Chapter 6	Representative checks establish tool behavior without pretending a full biological replication or unsupported walk.	Delete repeated generic explanations that movement alone is not verification; retain each experiment's actual assistance/interpretation limits. Add a compact appendix configuration map instead of another model-inventory essay.
Chapter 7	Useful synthesis and opportunities are present; existing workflow, sensory hierarchy, and hardware pipeline give it visual structure.	Shorten the long recap paragraphs at lines8/12/65/75. The last mixes several sensory and higher-control extensions. Use existing displays, shorter purposeful paragraphs, or a compact extension/question table; do not add decorative illustrations. Scope 'accurately enough to commit' to tested first-pass static sizing.
Torque paper	The physical capacity evidence is strong but arrives as new material in Discussion.	Put essential method definitions with Methods and measured comparisons with Results; reduce dense numeric paragraphs using the chapter's existing summary-table approach. This is new paper organization advice, not an instruction to override Ben's dissertation subsection arrangement.

An equation-rich appendix is not a wall-of-text failure. Long prose is a problem when it repeats, mixes purposes, or delays the evidence needed to understand it. The recommended first response is often deletion or consolidation.

Appendices and supplement

Material	Relevance / disposition
A, muscle equations	Keep. Necessary OpenSim/MuJoCo/BPA context and force/compliance equations. Resolve frame inconsistency and reduce code-history narration.
B, neurons/connectomes	Keep. Calibration, failed conversions, and actual W2L connectivity are valuable. Fix Figure B1 letters; remove GUI-acceptance bookkeeping from scientific narrative.
C, geometry/compliance	Keep authoritative coordinates, measured dimensions, held-out results, contact-release calculation, and 27-site survey. Repair contradictory frame authority and small route views; redundant bone view C2 is a deletion/crop candidate.
D, verification	Keep intervention protocols, convergence checks, IK/absolute-error evidence, assistance, and force comparisons. Repair score definition/captions and remove assembly/debug recaps that add no inference.
E, SADb	Keep physiological afferent/pathway tables; they fit and support the ME-to-neuroscience bridge. Replace the old screenshot; consolidate counts and use accurate note/source terminology.
Journal supplement	Keep relevant fit/coordinate/contact-release/survey evidence. Reconcile coordinate/tendon conventions, fix main-text inventory promises, enlarge labels, and remove an unnecessary forced page break.

No appendix is recommended for wholesale removal. More figures are not the remedy for every repetition. Every proposed deletion of scientific content needs a stated retained source and explicit disposition during any later editing pass.

Assistant scoring and learning

The requested rubric is in REVIEW_RUBRIC.md. Its largest failure cost is 40 for avoidable writing/evidence defects that invite an integrity challenge, followed by 30 for fabricated/unauthorized content, 25 for overriding or inventing a directive, 20 for missed substantive tracked/plain-text edits, 12 for an unanswered comprehension comment, and 10 for display/claim mismatches. It also penalizes substantive and copyediting tracked corrections separately. A reviewer insult is not itself evidence of misconduct, and an existing tracked-change record is not automatically a new unresolved failure.

Five documented historical calibration events have weighted cost 115. This is a relative assistant-failure ledger, not a thesis grade or a claim of statistical calibration. The system learns operationally through saved failure examples and regression checks: identify anchor, read reply/deletion, inspect effective source/figure, verify scope, compare all affected artifacts, and never claim completion from an old report. No model-weight retraining is claimed.

Detailed evidence and coverage

- TORQUE_AUDIT.md:15 findings; complete current paper/supplement source and rendered page review, Chapter 4 source and targeted figures.
- SADB_AUDIT.md:6 findings; full Chapter 5/AppendixE source and all their rendered pages. The AI-comparison item is removed entirely.
- SNS_DISCUSSION_AUDIT.md:11 findings; full Chapter 6/7/AppendixD source, actual included assets, and listed targeted rendered pages.
- APPENDICES_A_B_C_AUDIT.md:5 findings; all three sources and rendered pages193-249 reviewed for layout/relevance/internal consistency.
- REVIEW_RUBRIC.md:assistant penalty definitions, calibration examples, regression rules, and Ben's controlling exclusions.
- SOURCE_MANIFEST.json / ZIP_LOCAL_COMPARISON.json / PAGE_INVENTORY.json:source hashes, correspondence, text extraction, and page mapping. Build logs/QA remain under Archive/audit_20261030.

There are 37 detail findings, several describing the same repair across artifacts. Do not add their counts or weights as though they were independent scientific failures. The audit is not a reanalysis of raw measurements, a new literature replication, a plagiarism certification, or a complete live-history reconstruction. Published Chapter 20 was not audited or changed.

For a Frontiers submission, the primary journal check is conditional on the template's apparent destination: Frontiers Robot Design article types and author guidelines. No unsupported 15-item cap was imposed. The force-model publication record was checked at Actuators 15(5):230. These checks support the specific findings; they do not replace selection of the final journal by the authors.

Torque chapter, torque paper, and supplement audit

30 October 2026. Findings only; no manuscript, figure, result, parameter, or optimization was changed.

Scope and verdict

The source of truth is the latest supplied Overleaf ZIPs, isolated under `Dissertation/Archive/audit_20261030/dissertation` and `.../paper`. References below use ZIP-source line numbers, not older ProofFinal lines. The paper's current source was rebuilt by the root reviewer: 33-page main paper and 10-page supplement. I read both current source files, read Chapter 4 source, inspected all 33+10 rebuilt paper pages as contact sheets, and inspected the current biomimetic and physical-validation figures at larger scale. Chapter 4 detailed visual checks covered PDF pages 125, 127, 130-132, 134-135, and 137 (printed pages 104, 106, 109-111, 113-114, and 116). This is not an independent re-evaluation of every raw measurement or optimization output.

The mechanical contribution is substantial and identifiable: a force-to-joint model that includes effective compliance and wrapping, shared identification terms, independently routed paired actuators, and physical checks against specified single-muscle references. The current artifact is **not submission-cleared** because several definitions and records need reconciliation. This is a manuscript assessment, not an assessment of Ben's competence or integrity.

Provenance distinction: findings labeled **new audit recommendation** are reviewer recommendations generated in this audit. They are not represented as advisor requests. Previously recorded directives are explicitly identified. No FVU redefinition, refitting, additional optimization, measurement-uncertainty campaign, or waived individual vastus/pulley example is requested.

Ranked findings

ID	Priority	Evidence and implication	Minimal remedy and provenance
T01	P1, definition/provenance check	Paper `Main.tex:386` defines the required curve as 1.05 times the human target; `:600` reports minimum margins of +30.83% over required and +39.77% over plain human. Chapter 4 `30-torque.tex:709-722`, Table 4.11, repeats that pair. If both are ordinary relative torque margins on the same domain, they must satisfy `m_human=1.05(1+m_required)-1`; the first would imply 37.37%, not 39.77%. The unadjusted pair at paper :600 also does not obey this relation. This establishes an inconsistency of the written definitions, not which stored value is wrong.	Check the existing saved margin definitions, comparison targets, and angle domains. State those definitions or correct the specific transcription only after provenance is established. No new optimization. **New audit recommendation.**
T02	P1, equation consistency	Paper `Main.tex:176-194`, equations `eq:momentarm_def` and `eq:M_scalar`: `r=norm(d cross u)` is nonnegative, `F` is a magnitude, yet `Mz=rF` is introduced as a signed torque. The flexor data are negative. Chapter 4 :78-82 already uses `abs(Mk)=rF`, which avoids the contradiction. This is an exposition/sign-definition defect; it is not evidence that the plotted torque was computed incorrectly.	Use the magnitude form or define a signed moment arm and its axis convention explicitly. Retain the vector axis projection. **New audit recommendation.**
T03	P1, reproducibility check	Supplement `frontiers_SupplementaryMaterial.tex:69-89`, Table S2, supplies a biomimetic 10-mm flexor origin (-50,-45,32.8) mm and insertion (-27.9,-46,32.8) mm. Table S4 at :121-138 supplies the same named configuration with origin (-50,+35,50) mm and insertion (-12.24,-8.87,27.87) mm. The tables describe femur/tibia locations but do not give a transform or test-version mapping that reconciles them. Different reference origins or historical variants could explain this; neither set is silently declared wrong.	Identify which table defines each plotted experiment; state the frame transform or mark a historical configuration explicitly. Prefer one authoritative current coordinate inventory. **New audit recommendation.**
T04	P1 for distribution package	The source ZIP contains `Main.pdf` with 38 pages and a five-page supplement PDF, while rebuilding its `.tex` produces 33 and 10 pages. The bundled main PDF still reports old 66.32% hand-tuned coverage on p32 and shows the old Figure 8C; current source :655 gives 37.28→65.19% and the current two-panel figure. An uploaded generated PDF is distinct from Overleaf's current compile preview.	Replace or exclude the stale generated PDFs from any submission/share package and label the verified compiled outputs. Do not infer that the live Overleaf preview itself is stale. **New audit recommendation.**
T05	P2	Paper `BolenFrontiers22.bib:1085-1090` still calls the central force-model paper an unpublished manuscript submitted for publication; rebuilt paper p31 prints that description. The same ZIP's dissertation bibliography :1086-1095 correctly identifies *Actuators* 15(5):230, DOI 10.3390/act15050230.	Correct this bibliography record from the published source; no prose or force-model change is needed. **New audit recommendation**, independently verified at the publisher link below.

ID	Priority	Evidence and implication	Minimal remedy and provenance
T06	P2, carry-over gap	The dissertation's Figure 4.10 has the hybrid series removed and its legend outside the data. Current paper Figure 11 (rebuilt p22), source :410/:499/:505, still includes the approximate hybrid series and its discussion. Its inside legend also covers a large lower-left part of the data region.	Decide whether to carry the verified dissertation presentation into the paper; if so, update graphic, caption, and the three associated sentences together. This is a **carry-over recommendation from Ben's explicit Chapter 4 hybrid-removal/legend directive**, not proof that his earlier directive explicitly covered both manuscripts.
T07	P2, readability	Rebuilt paper Fig. 10 (p20) has legend text at about 5.39 pt, and Fig. 9 (p19) about 6.28 pt; supplementary Fig. S1 (p4) has legend text about 5.31 pt. These are PDF vector text sizes at the actual inclusion scale, not judgments based only on a thumbnail. Main Figure 13 also gives the extensor much less readable labels than the flexor. The repo calls for a 10-pt floor.	Re-export with larger fonts at final size, give plots more usable width, or consolidate redundant panels. Do not shrink a complete MATLAB figure and assume the native font size survives. **Standing figure rule plus new visual audit.**
T08	P2, organization	Paper :581-604 introduces configuration-specific objective equations and route-design settings inside Discussion; :604-626 introduces the physical tests and Figure 13 there as new results. The reader finishes Results without the most consequential hardware evidence.	Put essential cost definitions with Methods and the measured capacity comparisons with Results; retain interpretation and design trade-offs in Discussion. This recommendation applies to the **paper**; do not silently move Ben's specifically named dissertation section 4.3.4. **New audit recommendation.**
T09	P2, compression	Paper :382 is one long paragraph mixing both fit stages, test membership, optimizer settings, bounds, and initialization. Paper :596-602 then narrates route geometry, six release angles, old/new cam margins, pressure, clearance, and moment-arm statistics in dense paragraphs. Rebuilt pp25-26 remain heavily numeric prose. Chapter 4 already uses compact direct-route and cam summary tables instead.	Use a two-row fit-stage settings table and a compact route-summary table; retain a claim-first result sentence and interpretation. Move coordinate/release inventories to the supplement. Delete duplicated values from prose rather than adding another figure. **New audit recommendation**, consistent with Ben's prose-reduction direction.
T10	P2, parameter meaning	Paper :614 distinguishes installed flexor tendons of 28/30 mm from an effective 5-mm prediction parameter after fitting adjustment. Supplement :157 labels the redesign tendon 25 mm. These may be compatible conventions, but the supplement does not give the conversion explaining 25→5 or map all three quantities to the force-model length subtraction.	Add a short definition/equation distinguishing physical cable length, nominal routed length, and effective model parameter, with the existing fitting adjustment. Preserve the measured numbers. **New audit recommendation; clarification, not a claim the model inputs are wrong.**
T11	P2, author metadata	Dissertation publication note at `30-torque.tex:5` attributes the torque manuscript to seven named co-authors, including Elzein, Pang, and Scharzenberger; the paper author declaration at `Main.tex:48` and contributions at :696 list four. The titles also differ between the dissertation bibliographic record and current paper.	Ask the human authors to confirm the final author list/title and align the dissertation note and manuscript metadata. No authorship decision should be made by the agent. **New audit recommendation**, not an integrity allegation.
T12	P2, contribution framing	Paper abstract :81 promotes the first hardware measurements of the redesigned **extensor** only, while :612 and Figure 13 include fifteen redesigned-flexor measurements as well. The abstract's wording also repeats several technical qualifications but gives no concise measured-capacity outcome.	State that both redesigned routes were tested and give the existing all-15/all-9 measured-point result, scoped to the tested configurations. **New audit recommendation.**

ID	Priority	Evidence and implication	Minimal remedy and provenance
T13	P2, avoid algebraic repetition	Chapter 4 :140-265 develops many equivalent versions of Hooke's law, projected compliance, one-/two-bracket series compliance, and their component forms. Paper :245-318 retains a similar sequence. Core equations are valuable, but successive restatements consume reader attention without adding a new assumption or consequence each time.	Retain the stiffness assignment, projected compliance, series relation, and implicit force balance; make intermediate substitutions inline or put the full algebra in the existing technical appendix/supplement. Preserve the advisor-requested bracket development and its mechanics: this is **new compression advice**, not a request to remove it wholesale.
T14	P2, engineering figure labeling	Dissertation Figure 4.12, printed p110, shows coordinate ticks and p1…p9 labels but the included A/H panels lack x/y axis names and units. The caption does not supply the missing units. Its appendix version does label femur-frame x/y in meters. Closely clustered contact labels are also difficult to distinguish.	Restore frame/axis/unit labels on the main panels or state them in a concise caption; move contact labels slightly while keeping the geometry. **New audit recommendation.**
T15	P2, inventory accuracy	Paper `Main.tex:707-708` says Supplemental Data provides biomimetic fit statistics, but the supplement's fit table is the pre-optimization **pinned-flexor** table S1. Biomimetic statistics are in main Table 2.	Correct the supplement inventory sentence to list what is actually supplied; avoid promising a table that is not there. **New audit recommendation.**

What already works and should be preserved

- Chapter 4 section 4.3.4 has Ben's exact title, **Exceeding Human Torque Magnitude**. Figure 4.11 is a 1×2 plot and its panel-B legend is southwest; the earlier redesigned-route panel C is absent. This directly satisfies the recorded Ben directives.
- Chapter 4 Figure 4.10 correctly retains the apparatus and removes the hybrid scatter; the caption matches. This is stronger than the current paper presentation.
- The direct-route and cam tables materially reduce repetitive numeric prose. The physical measurements are distinguished from full-pressure predictions, and selected single-muscle references are named.
- The whole-leg coverage paragraph correctly states that the comparison mixes transmission, actuator sizing, and the baseline diameter substitution; it does not isolate placement. This avoids an exaggerated causal claim.
- The support-reaction discussion distinguishes the BPA origin mount from the fixed tackle mount and retains the direction-dependent force balance. The human-target versus pressure-specific-prediction distinction is explicit at the -86.5-degree measurement.
- The paper's current 33-page build has no visibly clipped main tables or missing major figures in the pages inspected. A clean compile is a technical success, not submission clearance.

Supplement relevance and visual disposition

Material	Relevance	Recommended disposition
S1 pre-optimization pinned-flexor fit table	Supports the choice of actuator-force model and direct comparison with hybrid calculations.	Keep; correct the main-text inventory.
S2 legacy attachment coordinates	Potentially valuable baseline reproducibility, but overlaps/conflicts with the current route inventory without a frame/version mapping.	Reconcile or explicitly designate historical; do not leave two unlabeled authorities.
S3 bracket references; S4 biomimetic 10-mm route; S5 redesign route	Necessary to reconstruct mechanics and compliance.	Keep, resolving T03/T10. A frame key is more valuable than additional explanatory prose.
S1 route comparison graphic	Gives direct/cam and human requirement context omitted from the main hardware figure.	Keep; enlarge labels and name the extensor target specifically rather than an ambiguous “vasti” legend if the intended target is vastus medialis.
S2 slippage photograph	Gives direct mechanical evidence of a changed force path.	Keep if linked to that interpretation; replace draft-like callout text with concise neutral labels when re-exporting.
Contact/release method and equations	Relevant to reproducibility and the cam/wrap algorithm; appropriate outside the main narrative.	Keep; the forced clear page before the survey leaves p7 largely empty (one short ending paragraph), so allow a better page break rather than adding content.
S6 27-site coordinate/size inventory	Necessary context for the aggregate whole-leg table.	Keep; consider an accompanying machine-readable table as an optional reproducibility improvement, not a new study.
S3/S4 eight cam poses and release legend	Relevant to checking the contact schedule, though visually repetitive.	Keep if each pose represents an event; otherwise use fewer key poses plus the existing release table. Increase contact-label clarity before adding detail.

Clearance criteria

These findings assess the submitted artifacts, not Ben's degree-level competence. I have not assigned chapter grades. Before journal submission, resolve the written margin and sign definitions, identify one authoritative mapping of coordinates and length parameters, ensure the circulated PDFs match the current source, and make the paper's plots legible at final size. The central mechanical contribution and discrete physical-capacity evidence are already present; the highest-value next pass is reconciliation and presentation, not repeating optimizations.

Primary external checks

The force-model paper is published as Bolen et al., Actuators 15(5):230, DOI 10.3390/act15050230; this supports T05.

The Frontiers Robotics and AI, Robot Design article-type guidance gives Original Research a 12,000-word maximum and the expected Methods/Results/Discussion structure. I did not certify an official count under its exclusions. No general 15-display-item cap was identified there for Original Research; the 15-item limit found for Registered Reports should not be applied to this paper's 13 figures and four tables.

The Frontiers author guidelines require ordered figure references, labeled axes with units, and clearly presented supplementary material. Exact target-journal/section choice remains a human submission decision; use these checks conditionally on Frontiers being the destination suggested by the template.

Chapter 5 and Appendix E audit

Reviewed the latest supplied Overleaf ZIP snapshot, extracted under `Archive/audit_20261030/dissertation`, not the older working chapter. No manuscript, app, database, or Overleaf review state was changed. Source line numbers below refer to that snapshot. Printed-page numbers refer to the fresh snapshot build at `Archive/audit_20261030/dissertation_build/main.pdf`.

Assessment and coverage

The resource chapter now has a clear purpose, a credible tools contribution, Methods/Results/Discussion organization, explicit category denominators, active curation language, and a useful bridge into neural-controller development. The evidence reviewed does not support an allegation of scientific misconduct or a judgment about the author's ability. The remaining problems concern presentation, traceability, and the distinction between source availability and curation status.

Read all 67 source lines of Chapter 5 and all 62 source lines of Appendix E, the 42-card historical review ledger and new Ben reply, app README and selected build/curation code, the local 2024 primary poster, and all rendered Chapter 5 pages (printed 119-128; PDF 140-149) plus all Appendix E pages (printed 282-285; PDF 303-306). Inspected the three included screenshots individually. Checked the existing app payload's aggregate retained/notes counts only; no classification review, new literature search, live database request, or reclassification was performed. Neuroscience citations in the appendix were not independently re-audited against every primary paper; no new physiological error is asserted here.

No P1 scientific-correctness blocker was established in this bounded audit. The highest priorities are P2 reader-comprehension and figure defects. P2 means substantive repair; P3 means optional polish.

Ranked findings

S1. P2: The advisor's table-screenshot objection remains visibly unanswered

- **Location:** `96-AppendixE.tex:18-20`, Figure E1, printed p. 283 / PDF p. 304; `50-database.tex:24` still sends the Table-tab explanation to the appendix.
- **Evidence:** Figure E1 still begins with LeCun's *Deep learning*, Hornik's universal-approximation paper, and later turbulence/deep-network papers. Several are explicitly tagged `app-only (removed from Airtable)`. The capture reports a 943-paper interface and lacks the current record-type and archive filters described in Chapter 5. This is the same screenshot criticized in historical advisor comment 5.23 (29 October, 10:11/10:13).
- **Why it matters:** A reader sees broadly unrelated archived material as the main demonstration of a curated sensory-afferent resource. The prose's 1,393-record deliverable and the pictured older interface are harder to reconcile. The existing archive explanation is helpful, but the demonstration still works against the chapter's purpose.
- **Minimal remedy:** Replace it with a real capture of the current Table tab, filtered to retained sensory-feedback records and showing a useful detail record; move that informative demonstration into the main chapter as the advisor requested. No papers need to be deleted or reclassified to do this. Preserve broader archive access and explain it once.
- **Origin:** Existing explicit advisor request, still open; high confidence.

S2. P2: “38 records without grounding text” conflates missing notes with missing sources

- **Location:** `50-database.tex:16` and `:63`; `96-AppendixE.tex:12`, printed pp. 120-121, 127, and 282.

- **Evidence:** Methods says records whose source text could not be located were removed from the working base. Appendix E then says, among the 561 retained papers, “38 records remain without grounding text.” The app payload verifies exactly 561 retained papers, 523 with notes, and 38 without notes. Its fields distinguish `n`/`nt` (notes) and `pdf`, but do not establish that those 38 lacked obtainable source text. The curation README also distinguishes not writing a note when text is unavailable from the wider question of whether text exists.
- **Why it matters:** The reader can reasonably infer incompatible selection rules, even if the underlying curation was sound. This recreates the source-availability confusion behind advisor comment 5.13, despite the newly improved active-voice paragraph.
- **Minimal remedy:** Identify what the 38 actually counts from the saved audit. If it is empty curation notes, call it that; do not equate it with absent source text. If these are a documented older snapshot or explicit exceptions to the removal rule, state that scope briefly. Also distinguish archived-only missing-source records from retained working-base records in the Further Improvements sentence. No new corpus-wide review is needed merely to make the counting language accurate.
- **Origin:** New cross-section consistency finding, informed by an existing advisor concern. This is a wording/provenance ambiguity, not evidence of fabricated notes. High confidence about the counted missing-note field; source-availability status remains unverified.

S3. P2: The pivot screenshot is effectively unreadable at its final inclusion size

- **Location:** `50-database.tex:41-43`, Figure 5.2, printed p. 125 / PDF p. 146.
- **Evidence:** The 1800-by-1100 raster devotes most of its lower half to an empty browser canvas. The many animal-category columns and small cell counts are reduced to very small type at `0.96\textwidth`. The compiled figure occupies a largely blank float page, while the useful top portion is only a narrow strip. The source capture also includes a rebuild command and old 943-paper header that contribute nothing to the argument.
- **Why it matters:** The illustration cannot practically substantiate the paragraph explaining cell counts and drill-through. More pixels alone will not fix a table with too many columns for a portrait text width.
- **Minimal remedy:** Crop the unused canvas and interface clutter, and show a deliberate, legible subset of columns/rows or use a landscape presentation. Alternatively replace the static demonstration with a native table of selected pivot cells and a short note about interactive drill-through. Retain the multi-tag counting caveat. Do not manufacture new counts.
- **Origin:** New visual audit finding; high confidence, checked at full compiled-page scale.

S4. P2: The six bubble-map callouts are readable, but the evidence they point to is too small

- **Location:** `50-database.tex:26-35`, Figure 5.1, printed p. 124 / PDF p. 145.
- **Evidence:** A-F labels and leaders are clear. The actual notes, record metadata, connection list, and map controls are extremely small in the compiled sideways figure. The reader is asked to inspect curation and translation content that is not comfortably legible. The capture also retains the older paper-only header.
- **Why it matters:** The schematic organization is successful, but screenshot content cannot fully support the detailed prose walkthrough. This is a final-size legibility problem, not a reason to discard the six-callout design.
- **Minimal remedy:** Keep A-F, enlarge the screenshot content, and use selective detail crops for the left notes and right connection list within a revised composite if necessary. Remove decorative interface chrome. Check the full page at actual print size before accepting it.
- **Origin:** New visual audit recommendation; high confidence. The existing requirement to call these callouts rather than panels is already satisfied and should be preserved.

S5. P2: Coverage statistics are repeated, and the first visual arrives after five text pages

- **Location:** `50-database.tex:8,20,22,37,63`; `96-AppendixE.tex:8-14`; printed pp. 119-123 and 282-283.
- **Evidence:** The 1,393/943 split, coverage counts, and graph counts recur in the opening, Results, Pivot explanation, limitations, and appendix. The appendix repeats the full-set coverage paragraph before adding the retained-set counts. Printed pp. 119-123 are all text before Figure 5.1 appears on p. 124. The map paragraph at line 26 is approximately a page of operational description, followed by another connection-list walkthrough.
- **Why it matters:** The chapter asks the reader to retain many counts and interface details before seeing the app. It has a local wall-of-text problem, even though individual paragraphs are generally grammatical. This is not a demand to add a figure to every section.
- **Minimal remedy:** Consolidate full-versus-retained coverage into one native table, probably in Appendix E, with explicit denominators. Keep headline scope counts in the chapter and avoid repeating the whole field inventory. Bring the existing app demonstration closer to its first substantive discussion. Trim repeated navigation instructions and preserve the engineering use case.
- **Origin:** New flow/compression recommendation, consistent with the advisor's earlier organization and topic-sentence comments; high confidence.

S6. P2: The resource is described, but its retrievable release is not identified

- **Location:** `50-database.tex:8-10,20-22`; `96-AppendixE.tex:4-14`.
- **Evidence:** Neither chapter supplies an application download/repository path, a version or snapshot date, or a persistent release identifier. The bibliography cites the 2023 and 2024 posters, whose URLs point to the project/conference rather than the actual app. Local code shows that records, citation graph, and clusters are embedded at build time; consequently the specific snapshot matters.
- **Why it matters:** A dissertation reader cannot readily obtain the 1,393-record artifact or know which saved data produced these counts and diagrams. This weakens the reproducibility and practical value of a resource contribution.
- **Minimal remedy:** Add one compact availability statement with the verified app/repository location and archived snapshot/release identity. State the curation/citation snapshot date if it is recoverable. Do not put build commands from the screenshot into the reader's main flow.
- **Origin:** New audit recommendation; high confidence that this information is absent from the two audited sources. No live hosting or publication is requested by this report.

AI-comparison scope ruling

Ben clarified: “There's no documented assessment method. It was my judgement, compared to how well and fast I can do things.” The AI comparison is excluded from findings and penalties. No further check, wording revision, method, rubric, benchmark, or study is assigned. The earlier inferred “requested AI-rating methodology” task is withdrawn.

Appendix relevance and tables

Appendix E is relevant. Its two physiological tables provide the signal-to-pathway bridge that a mechanical-engineering reader needs; they should be retained rather than cut merely to shorten the thesis. Table E1 (printed p. 284) and landscape Table E2 (p. 285) fit their pages, have visible headers/citations, and do not exhibit clipping in this build. Table E1's III/IV signal cell contains a long explanation of convergence that would fit more naturally in the pathway-role column, but that is P3 column organization, not an independently verified scientific error.

The principal appendix repairs are replacing Figure E1, distinguishing note coverage from source availability, and reducing duplicate coverage prose. The 1,000-record Airtable limit at `96-AppendixE.tex:8` explains the pressure to

archive, while Chapter 5 explains scope/source selection; one short distinction between storage limit and selection criteria would prevent the impression that all 436 archived records were excluded for one scientific reason. Existing counts should not be silently changed.

Advisor comments that are already addressed in the ZIP

- Four main tabs are named and explained before detailed descriptions.
- Offline functionality is grounded in embedded data; OpenAlex online search is identified separately.
- Duplicate Zotero records and backward/forward citation searching are explained.
- Ben's new reply is reflected in active curation language and retention/removal criteria.
- Table, map, and Pivot descriptions are organized more consistently.
- Citation directions are stated as papers citing the focus versus papers cited by it.
- A-F are called callouts, and each marked feature receives an explicit figure reference.
- Both presentation citations are at the chapter opening.
- Discussion explains learning the literature and choosing neural-model pathways rather than merely re-listing features.
- Sparse classification is acknowledged; the larger classification exercise remains deferred by Ben and is not reopened here.

Historical cards are evidence of the advisor's concerns, not proof of current live comment status. The ZIP contains source and figures, not the current Overleaf review database. This report does not claim any thread was resolved or any tracked change was accepted.

Chapters 6-7 and Appendix D audit

Audit date: 30 October 2026. Findings only; no manuscript, figure, model, or numerical changes.

Source and coverage

Authority is the dissertation ZIP downloaded 29 October at 23:52, extracted without changing the working source into `Dissertation/Archive/audit_20261030/dissertation`. All line references below are to that snapshot's `chapters/60-sns.tex`, `70-overall.tex`, or `95-AppendixD.tex`, not to older local drafts. The freshly compiled snapshot is `Archive/audit_20261030/dissertation_build/main.pdf` (306 PDF pages). Printed page numbers differ by 21 from PDF page indexes in this build.

Read all three source files. Visually reviewed the included assets for the workflow, library/neuron implementation, knee transient, loaded arm, W2L feedback comparison, IK overlay, BPA construction/cutaway, first-pass design workflow, control hierarchy, hardware pipeline, spiking knee comparison, and force comparison. Inspected compiled pages 140-145, 151, 154-156, 163, 166, 168, 253, 265, 267, and 281, with full-sized source-asset checks supplementing the page contacts. Other cited source passages were read; no claim of visual inspection of every page. QA images are under `Archive/audit_20261030/sns_qa`.

Context read: dissertation skill and relevant AGENTS context; consolidation and flow/validation reports dated 28 October; the current source's experimental-support qualifications. Current executable scoring code was read solely to identify a source-version question. It is not assumed to be the exact historical code used for every archived table. No new simulation, fit, optimization, or literature-replication result is asserted.

All recommendations below are **NEW AUDIT RECOMMENDATIONS**, not quotations of advisor requests or already-authorized editorial changes. The parent's advisor-comment reconciliation is separate.

Overall assessment

These chapters contain defensible doctoral engineering work: matched inputs, no-stimulus controls, pre-intervention equality checks, layer-specific interventions, explicit assistance accounting, and separation of implementation agreement from biological equivalence. The absence of harness-free walking is stated directly. A tools contribution does not require pretending that walking has been solved. The main risks are an inadequately specified scoring metric, ambiguity among numerous model/configuration families, a small number of inaccurate captions, and overextended summary prose.

The highest-value response is clarification and compression of existing evidence. It is not a larger set of simulations or a figure added to every paragraph.

Ranked findings

S01 - P1: identify the exact gait-score definition behind the reported numbers

Evidence: D:125 says “The cycle-shape metric assigns zero to perfect agreement”; D:133 and captions at D:136 and D:220 use the same interpretation. D:155 calls tuning errors “mean-removed cycle-shape errors.” Yet no operational equation, component weights, or score-version identifier appears in the chapter/appendix sources. The tables report values from approximately −145 to −10,040, which a reader cannot interpret quantitatively from “cycle shape” alone.

Source trace requiring reconciliation: current repository `Code/MuJoCo_SNS/spinal/kine_ref.py:271-275` includes an absolute mean-angle error term, and lines 320-331 apply a bilateral bonus of 40 before returning minus total cost. Thus the current full function is not simply a mean-removed error with an invariant perfect score of zero. This does **not** establish that every archived table used that exact revision. The missing connection between archived results and the score definition is itself the problem. Code comments also retain older “zero perfect” language and should not

substitute for checking executable operations.

Minimal remedy: identify the archived scorer/version used for each result family; give its equation or a short term/weight table, define the failure sentinel separately, and distinguish the mean-removed shape component from the full objective. Preserve reported values unless their provenance establishes a specific error. No new tuning or simulation is needed to document an existing score. **Confidence:** high that the manuscript definition is incomplete; exact historical metric mismatch remains conditional on archived-version checking.

S02 - P2: Table D5's contact-duty caption contradicts its zero-cycle row

Evidence: D:220, Table `tab:spike_ground` (printed p.267), states “Knee minima and contact duty are computed over detected cycles.” D:227 reports 9/0 right/left cycles, with the left frozen; D:229 nevertheless gives left contact duty 0.985. A cycle-conditioned left-leg duty has no detected cycles on which to operate.

Minimal remedy: check the saved diagnostic definition and label duty as the time fraction over the scored window if that is what it is; state how the single reported knee minimum is selected. Do not recalculate the simulations. **Confidence:** high caption inconsistency; exact underlying duty denominator needs verification.

S03 - P2: distinguish the old spiking startup comparison from the new settled transient experiment

Evidence: D:246-269 call the five-second knee comparison a “matched operating point” and its final two seconds “settle statistics”; the included `simulink_knee_spiking_twin.pdf` still shows increasing knee angle over those final two seconds and uses positive angles. By contrast, D:60-64 explicitly says the original settings did not settle, introduces a changed ramp-width configuration, runs 50 seconds, and reports a flexion-negative baseline. Chapter 6's Figure 6.4 uses the new settled sensory-current experiment.

Why it matters: the existing comparison can remain useful as a graded/spiking startup response, but a reader can mistake it for a spiking replication of the newly validated transient test or for a demonstrated steady operating point.

Minimal remedy: identify it as the earlier startup configuration, rename “settle” to “final-window” statistics, and state its coordinate convention and relationship to the modified transient configuration. No demand for a new spiking experiment. **Confidence:** high from source and actual traces.

S04 - P2: the first-pass workflow omits the second acceptance gate described in prose

Evidence: 7:20 states “Failure at either check returns the design to actuator selection or route revision.” Figure `fig:design_workflow` (`design_workflow.pdf`, Figure 7.1) has one “Meets targets?” decision before “Closed-loop verification,” then a direct arrow from that box to “Commit to hardware.” There is no closed-loop pass/fail branch or return route.

Minimal remedy: make the existing figure match the stated two-check process, or narrow the prose. A failed closed-loop simulation must not visually lead straight to hardware commitment. **Confidence:** high.

S05 - P2: several diagrams lose the labels needed to inspect the claimed implementation

Evidence: 6:102-104, Figure 6.3, printed p.140: the native-neuron diagram has substantially smaller block/port labels than the body text at final inclusion size. 6:153-155, Figure 6.4, printed p.144: the knee illustration in B is tiny within a large surrounding space; its BPA and knee-angle labels are difficult to read. The source knee layout confirms that the small schematic is embedded in an otherwise much wider page, so simply increasing total figure width will not solve its internal scale. The full-sized source arm and transient curves are clear; the weakness is allocation of area and inconsistent label size.

Minimal remedy: crop unused internal space and enlarge B within the existing figure; use a simplified inspectable neuron signal-flow diagram or a readable crop for Figure 6.3B. Preserve the library's twelve-block overview and six feedback connections. **Confidence:** high visual finding; no numeric font-size threshold was inferred without measurement.

S06 - P2: Chapter 7 has the largest remaining walls of text, and new figures are not the best cure

Evidence: 7:8, 7:12, 7:65, and 7:75 are each single source paragraphs containing roughly 200-285 regex-counted word tokens including LaTeX arguments. In particular 7:75 combines vestibular sensing, visual correction, cerebellar adaptation, basal-ganglia selection, multiple precedents, and database coverage. The sensory-control hierarchy already exists as Figure 7.3, so another figure would duplicate its function. The paragraph at 7:8 mixes the main thesis outcome, all four correction terms, a validation statistic, and speculative $\chi 4$; 7:12 mixes prior work, final knee outcomes, general multi-joint routing, and cam design.

Minimal remedy: retain the synthesis claim and scientific implication in the main discussion; cut the parameter/coverage recaps already supplied in Chapters 4-6; make the four sensory extensions short distinct paragraphs or a compact extension/question/evaluation table using the existing hierarchy as the organizing graphic. This is primarily deletion and restructuring, not an image shortage. **Confidence:** high; the rendered Chapter 7 opening confirms the dense presentation.

S07 - P2: consolidate repeated explanations of why movement is not verification

Evidence: 6:4, 6:16, 6:26-30, 6:34-38, 6:67, 6:137-139, and 6:234-244 repeatedly frame the same distinction among a moving mechanism, an isolated rhythm, successful interface exchange, and verified coupled behavior. The actual tests later do the necessary evidential work. Chapter 7:16 and 7:65 recap the same construction/interface/curriculum sequence again.

Minimal remedy: retain the causal motivation once in the MATLAB-attempt account, define the verification questions once before Results, and let the individual experiments answer them. Keep configuration-specific limitations where they affect interpretation; delete repeated generic warnings and tool inventories. The two chapters need synthesis at the end, not another list of components. **Confidence:** high editorial finding; does not propose removing assistance caveats.

S08 - P2: add a compact configuration map instead of making the reader reconstruct model identity

Evidence: 6:85 introduces human Gait2392, its BPA counterpart, Li-Deng/W2L, CAD-derived mechanisms, full/pruned/six-synergy/W2L-layout neural variants, and spiking counterparts. D:298-360 introduces 411 N body weight, 75 kg CAD mechanics, different support fractions, separate replay and drive-weaning configurations, 262- and 544-neuron CAD organizations, and reference-tension rather than calibrated BPA actuators. Existing Table 6.1 lists capabilities, not which result belongs to which actual model/controller/support combination.

Minimal remedy: replace part of the repeated model-inventory prose with one appendix table: model/mechanics, controller ID, actuator law, assistance, evaluated quantity, saved configuration identifier, figure/table reference. Keep the main chapter focused on the few representative comparisons. This protects against readers interpreting distinct support percentages as a matched support-reduction experiment. **Confidence:** high organization/reproducibility gap; not a claim that any numerical result is false.

S09 - P2: qualify the scope of the opening “accurately enough to commit” claim

Evidence: 7:8 says “a designer can predict actuator force, joint torque, and range of motion accurately enough to commit to an actuator set, a resting length, and a route before the hardware exists.” It then discusses configuration-dependent residual errors. In the same chapter, 7:53 says physical clearance, load sharing, and off-axis verification remain necessary. Chapter 6's dynamic models omit pneumatic transients, while the experimentally supported torque envelope is isometric.

Minimal remedy: scope the contribution to first-pass static sizing and route selection within the tested configurations, with mechanical feasibility and dynamic behavior as subsequent checks. Preserve the positive design contribution; avoid a universal prospective accuracy guarantee for a full leg. **Confidence:** high wording exceeds the immediately documented scope; not an instruction to add Ben's rejected measurement-uncertainty task.

S10 - P2: make Figure 6.8 show its central coordination result directly

Evidence: 6:209's central result is a contact-phase change from 0.498 to 0.223 cycles against a 0.35-0.65 criterion. Figure 6.8 (`w2l_feedback_verified.pdf`, printed p.151) shows dense 25-second knee/contact traces and displacement, but has no phase annotation or criterion band. Its two force axes also use different limits, reducing direct amplitude comparison. The caption declares the condition fails a phase criterion without showing the measured phase.

Minimal remedy: annotate each existing column with its measured median phase and the acceptance interval; use common force/displacement limits for a comparison that is meant to be visually matched. A short matching time-window inset is optional, not necessary if the numerical annotation suffices. Do not rerun or retune. **Confidence:** high visual/argument finding.

S11 - P3: remove implementation bookkeeping from the scientific narrative and tighten table language

Evidence: D:358 explains that “four named ankle indices alias knee-PF cells and are not additional neurons.” That implementation distinction matters to a code audit, but no scientific inference in the surrounding paragraph relies on the alias names. D:285 labels a failed spiking conversion “dead”; D:231 mixes “no / up” under “Non-finite states / fell.” D:302 uses the unexplained term “zero hand drive.”

Minimal remedy: retain correct model census and controller authority, but keep alias/debug history in the linked technical report; use “no sustained alternation,” separate yes/no fields, and “zero externally supplied neural drive.” **Confidence:** high prose precision finding.

Appendix D relevance and display quality

Retain the pulse protocol and numerical convergence checks, layer-stimulation controls and traces, explicit absolute-angle errors, pathway-deletion table, matched support accounting, and pose-wise force comparison. These are directly relevant evidence for Chapter 6's claims. The IK overlay and the explicit distinction between squared correlation and R^2 are strong safeguards.

Compression candidates: Figure D3B repeats the knee illustration already in Figure 6.4; historical assembly/rerun descriptions can be reduced when they establish no new inference; alias/debug narration belongs in the technical repository records. Do not remove a negative or a support caveat simply to make the work appear more successful. The normalized force figure is clearly labeled as curve-normalized and the absolute agreement table is separate, so normalization is not being hidden; the paragraph correctly says lag cannot establish tendon-compliance delay.

No visually established clipping or table overflow was found on the inspected Appendix D pages. Tables D5/D6 fit, but their captions need the corrections above. Figure D10's compact six plots are readable at enlarged PDF view; preserving vector output and checking final print size remain advisable.

Positive findings to preserve

- Correctly distinguishes the reduced neural-only Rybak check, the Deng-derived suspended unilateral test, and full Gait2392 feedback experiments.
- States that a motor phase shift alone does not prove rhythm-generator resetting, and includes no-deletion controls.
- States that removing neural drive differs from removing mechanical assistance, that the CAD allocator cancels ordinary PF/reflex authority, and that no harness-free walking was demonstrated.
- Reports negative force R^2 values and large IK angle errors rather than hiding them behind correlations or scalar scores.
- Distinguishes neural-backend acceleration from whole-simulation speed and deterministic replay from robustness.

- Labels the control hierarchy and hardware pipeline additions as proposed work; existing figures supply useful synthesis without requiring new illustrations.

No revision was authorized or performed by this subagent. The P1 metric item is a documentation/provenance priority, not a claim of fabrication and not a request to rerun optimization.

Appendices A-C audit

30 October 2026. Findings only. References are to the isolated latest Overleaf source snapshot in `Archive/audit_20261030/dissertation/chapters`, not an earlier working tree. Printed pages are used below; add 21 for the fresh PDF's page index.

Read Appendices A, B, and C in full. Reviewed all rendered pages 193-249 in contact sheets and read the source and pertinent figure labels. This supports a layout/relevance and internal-consistency audit, not certification of every literature equation or executable evaluator against raw data. No source, figure, result, or parameter was changed.

A1. P1: contradictory descriptions of the identified bracket-frame convention

Evidence: `94-AppendixC.tex:186` identifies the dated two-bracket/two-transform result and states that both pinned-flexor brackets use the two-rotation construction. At `:216`, it instead says both use the pitch-only assembly; Table C9 at `:226-227` labels both rows pitch-only. `92-AppendixA.tex:198` also calls both frames pitch-only. These passages all present themselves as the implementation of record. The contradiction is visible on printed pp.201-202 and 227-229. Appendix C itself states that the origin stiffness array does not possess the rotation invariance claimed for the insertion array, so this distinction cannot simply be dismissed as redundant wording.

Minimal remedy: distinguish the saved identification configuration from the present evaluator implementation, using the existing dated result/source record. Align Appendix A, Appendix C prose, Table C9, and the chapter's description. This is a documentation reconciliation. No coefficient alteration, re-fit, or new optimization is justified by the textual contradiction alone. **Origin:** new internal-consistency finding. **Confidence:** high contradiction; exact authoritative implementation requires the existing saved-run provenance, not a guessed choice of convention.

A2. P2: Figure B1's caption assigns six letters to a three-condition labeling scheme

Evidence: `93-AppendixB.tex:190`, printed p.215, describes graded baseline as (A,B), full spiking conversion as (C,D), and threshold-only conversion as (E,F). The actual graphic labels its three rows/conditions (A), (B), and (C); the right-column plot of each row has no separate B/D/F letter. Thus the caption can send a reader to the wrong condition.

Minimal remedy: describe the existing row labels correctly: (A) graded baseline, (B) full conversion, and (C) thresholds only; identify hip and knee by columns. Alternatively reletter the six plots consistently. Preserve all traces. **Origin:** new visual/caption finding. **Confidence:** high.

A3. P2: some Appendix C geometry figures cannot communicate their intended detail at print size

Evidence: Figure C1, printed p.234 (`94-AppendixC.tex:322-325`), shrinks eight tall route views into one portrait-width figure; contact labels and the legend are very small and several contact names overlap. Figure C2, p.235 (`:330-333`), uses a broad 3D axis box with the bone cloud compressed into a very small central area; it adds little inspectable anatomical context relative to the page area used. Figures C3/C4, pp.242-243, have small multi-axis legends/ticks, while the later C5/C6 pair allocates space more effectively.

Minimal remedy: use C5/C6's event-pose layout as the better model; crop/zoom C2 to the relevant knee geometry or delete it if C1/C5/C6 already establish the same point. Reduce duplicated pose inventories before adding more figures. Enlarge meaningful labels at the final inclusion size. Do not replace measured results or route coordinates with illustrative geometry. **Origin:** new visual/relevance recommendation. **Confidence:** high legibility/space-allocation problem; no unmeasured font-size value is asserted.

A4. P2: Appendix C repeats model history and constants rather than providing one compact specification

Evidence: `94-AppendixC.tex:11-20,99-105,151,155,186,214-216,234-255` alternates test inventories, saved-model parameters, evaluator history, frame choices, and embedded file/line provenance. The long joint-transform paragraph at :155 packs coordinate ranges, transform composition, spline details, and multiple implementation pointers into one paragraph; C.1-C.5 repeatedly explain which tests are training, validation, or unused despite the tables already carrying those fields. At :387-391, the appendix returns to a route-builder debugging history and concatenation failure rather than a compact supported-result specification.

Minimal remedy: retain the equations, per-test/actuator tables, exclusions, frame definitions, model-input versus measurement distinction, and contact-release algorithm. Consolidate the configuration-to-source mapping once, then use short cross-references. Move debugging chronology and superseded array stories to repository technical notes. A small frame/quantity table is more useful than another explanatory essay. **Origin:** new compression recommendation, consistent with Ben's rule against corrective-action narratives. Do not restore his rejected bracket-shift story or remove negative results.

A5. P3: remove editorial acceptance bookkeeping from Appendix B

Evidence: `93-AppendixB.tex:194` says graphical verification or an equivalent acceptance ruling remains outstanding because the headless artifacts have not been opened in the AnimatLab graphical environment. This is a project-workflow status, rather than an additional numerical result or a defined scientific acceptance criterion. At :148, several 17-digit baseline scores also occupy scientific table space primarily to express a software identity test.

Minimal remedy: retain the headless-test scope and explicit failed dynamics; keep GUI task status and full identity strings in the archived verification report. In the appendix, state the identity-test result and give only the precision needed for the scientific comparison. This is optional presentation compression; no value should be silently changed. **Origin:** new editorial recommendation.

Relevance and retention decisions

Appendix	Assessment	Keep	Compress or repair
A: force/torque equations	Relevant and technically useful for an ME reader moving between OpenSim, MuJoCo, and BPA mechanics. Its equation-rich pages are not a wall-of-text defect.	Constitutive equations, sign definitions, human target construction, compliance/force balance, and units.	Resolve A1; shorten implementation archaeology and overbroad opening claim that this traces any number in the dissertation.
B: neurons/synapses	Relevant; separates archived 410-neuron comparisons from the independently replayed W2L network and reports failed conversions. Tables B1-B5 fit their pages; W2L connectivity/conductance figures are appropriate.	Neuron/synapse definitions, calibration factors, pass/fail gates, and verified connectome.	Fix Figure B1 caption; compress historical debugging and acceptance bookkeeping.
C: geometry/compliance	Essential reproducibility support. Coordinate and per-actuator dimensions tables belong here. Contact/release equations justify the algorithm and should not be cut merely to save pages.	All authoritative test/configuration tables, full held-out results, measured dimensions, contact-release method, and 27-site inventory/coverage.	Resolve the frame contradiction and duplicate inventories; improve C1-C4 or remove a redundant view.

Assistant review rubric and failure costs

30 October 2026. This rubric evaluates the assistant's editing/review failures. It does not grade Ben, assign a PhD score, score his AI-tool comparison, or treat a reviewer's tone as evidence of scientific misconduct. It is a persistent error-analysis/checklist system; no model weights were retrained.

Weighted penalties

The objective is zero avoidable misses. Report the number of confirmed unique failures and their weighted cost, not a spurious percentage probability of acceptance. Costs can exceed 100 across a campaign. Count one underlying miss once at its highest applicable cost; record associated comments as evidence rather than adding duplicate penalties. Do not count a comment as unresolved merely because its review thread is still open.

Cost	Assistant failure	Trigger and required response
40	Wording/evidence failure that needlessly exposes the author to an integrity challenge	Verify the actual trigger. Remove unnecessary self-defense, ungrounded claims, or contradictory selection rules; preserve accurate methods and the author's reply. A hostile comment alone proves no scientific fault.
30	Fabricated source, result, completion claim, or unauthorized scientific change	Stop clearance; recover original evidence and disclose the exact error. No silent harmonization.
25	Misreading or overriding Ben's directive; presenting an assistant inference as an advisor requirement	Record instruction provenance verbatim. Withdraw the invented task; do not replace it with another unsolicited task.
20	Missed substantive tracked edit, resurrected deletion, or failure to read an author's plain-text changes	Capture author, changed text, anchor, and date; reconcile against the effective source. Metadata absence is not evidence of absent edits.
12	Unanswered 'huh?', 'which?', 'what do you mean?'	Locate the actual anchor and repair the concept/referent. If the anchor is uncertain, say so instead of declaring success on a nearby paragraph.
10	Figure/table/claim mismatch, illegible evidence, wrong caption, or broken cross-reference	Inspect the final compiled size and content; check units, letters, denominators, and consistency with the claim.
8	A substantive advisor tracked replacement that an adequate editing pass should have caught	Treat as a labeled example of a recurring reader problem; learn the principle, not merely the replacement string. Mere use of track changes is not an automatic defect.
6	Repeated result inventories, generic caveats, operational walkthroughs, or long unfocused paragraphs	Prefer deletion, consolidation, or an existing display. Do not add a decorative figure to satisfy a quota.
2	Copyediting-only tracked correction	Record the recurring grammar/notation rule and apply it consistently without broad rewriting.

Historical calibration and regression rules

Evidence	Assistant failure / cost	Prevention rule
Chapter 5 comment5.13, 29 Oct09:40; Ben reply 14:57	Integrity-inviting curation narration, category40. The original wording invited a gratuitous inference about the author rather than describing selection.	State the true active curation rules; preserve Ben's reply; check cross-section denominators/status definitions. No accusation is inferred from tone.
Ben's saved Word anchors, 29 Oct16:14	Treating absent Word comments/tracking as absent new text, category20.	Read the complete current document text and compare content, including headings and Disposition fields.
Advisor tracked deletions 08:47/08:49, Chapter 4	Missing effective deletions in a source capture, category20.	Read tracked edits separately from comments; check that deleted passages stay deleted in effective output.
Chapter 5 comment5.23 screenshot	Calling the earlier polish complete while the criticized screenshot remained, category10.	Verify the included asset itself at final size, not just revised prose.
This audit's initial 'documented assessment method' recommendation, corrected by Ben 30 Oct	Misattributing an inferred task to the advisor, category25.	Exclude the AI-comparison issue entirely. It was Ben's judgment of accuracy and speed relative to his own work. No assessment-method, attribution-check, benchmark, or wording task remains.

These five documented calibration events have weighted cost 115, counted once each. This is a historical sample used to set priorities, not a score for this manuscript or an exhaustive failure count. Current live tracked-change totals were not obtained and are not invented.

Acceptance checks before saying an edit is done

1. Identify authority: newest supplied ZIP, advisor thread/anchor, Ben reply, Ben directive, or new recommendation. Keep these distinct.

2. Read the effective source and actual included figure. A saved comment or old report is not current-state proof.

3. Trace each changed number, equation, model identity, and claim to the existing record. Resolve contradictions before stylistic rewrites.

4. State the engineering result before its display when appropriate; define the target and the conditions under which it holds.

5. Assign every paragraph one purpose. Remove duplicate recap and operational detail before adding graphics.

6. Check final-size axes, units, legends, letters, captions, table denominators, and first callouts. Equations belong in text unless naturally part of a schematic.

7. Compare article/dissertation/supplement carryovers explicitly. Never assume a change to one changed the others.

8. Build the reviewed snapshot and inspect affected pages. A clean build is necessary, but does not clear scientific or layout findings.

9. Report addressed / partial / pending / deferred-by-Ben separately, with evidence. Do not accept tracked changes, resolve threads, or upload during a findings-only audit.

Boundaries retained

FVU is settled by Ben's reply: no refit, metric replacement, or optimization rerun. The database classification exercise remains deferred. Published Chapter 20 is protected. The waived individual vastus/reverse-pulley worked example is not revived. True assistance conditions and negative results remain visible; administrative debugging history need not remain in the scientific narrative.

Use `SOURCE_MANIFEST.json`, the three subagent reports, and `APPENDICES_A_B_C_AUDIT.md` as the labeled examples for the next pass. Add a regression rule only when an observed failure justifies it; do not manufacture new requirements from every suggestion.